Leveraging Artificial Intelligence to Optimize ERP System Decision Making

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1 Abstract

Enterprise Resource Planning systems suffer from heterogeneous data schemas across vendors (SAP, Odoo, Oracle NetSuite, Microsoft Dynamics 365, Epicor, Infor). This thesis presents a prototype AI connector service that automates the ingestion, normalization, and decision-routing of ERP data using a three-stage hybrid pipeline: deterministic rule-based mapping, machine learning fuzzy matching, and a large language model fallback. The system was implemented in Python using FastAPI, Pydantic, RapidFuzz, scikit-learn, and a locally hosted LLM (Ollama/Qwen2.5-Coder), and integrated with a Spring Boot ERP backend. Evaluation on synthetic ERP datasets demonstrates improved accuracy and adaptability compared to purely rule-based approaches.

2 Introduction

2.1 Overlying Question and Objectives

Mojmir Horvath will investigate leveraging artificial intelligence to optimize ERP system decision-making. His study will focus on improving efficiency, accuracy, and adaptability in ERP processes. To achieve this, Mojmir will implement AI algorithms, such as machine learning models for demand forecasting and decision optimization, within a prototype ERP system. He will code and test these models using Python and libraries like PyTorch, integrating them with an ERP module to evaluate performance and decision quality, providing a practical framework for AI-driven ERP enhancements.

3 Background and Related Work

3.1 ERP Systems Overview

- Overview of major vendors: SAP S/4HANA, Odoo, Oracle NetSuite, Dynamics 365, Epicor, Infor.

3.2 Data Integration Challenges

- Schema heterogeneity across different ERP ecosystems.

- Field naming inconsistency and lack of standardization.
- Complex nested data structures.

3.3 AI in ERP Applications

- Current trends: demand forecasting, anomaly detection, and intelligent routing.

4 Theoretical Foundations

4.1 What is AI?

Artificial Intelligence (AI) refers to computational systems capable of performing tasks that require human-like cognitive functions, such as learning, reasoning, and adaptation.[1] Within enterprise software, AI encompasses the ability to extract meaning from business data, infer relationships between heterogeneous schemas, and support decision-making through predictive or generative methods. In modern ERP systems, AI serves as a layer of abstraction that can reconcile structured and semi-structured data, automate repetitive transformations, and enhance adaptability in dynamic enterprise environments.[2] This research applies AI as an integrative mechanism to align inconsistent data structures between ERP systems, enabling rule-to-learning transitions when deterministic approaches fail.

4.2 Relevant Techniques for Data Harmonization

Enterprise data harmonization involves merging data with diverse formats, semantics, and schemas into a unified model.[3] Multiple paradigms underpin this process:

- **Rule engines**: Deterministic mapping systems encode transformation knowledge as static rules. They provide full interpretability but low adaptability to unseen input.[4]

1. Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach* (Prentice Hall, 2010).
2. Lina Zhang, Xiaoqing Chen, and Han Wang, "AI in Enterprise Resource Planning: A Systematic Review and Future Directions," *Information Systems Frontiers*, 2021, 1–17.
3. Anhai Doan, Alon Halevy, and Zachary Ives, *Principles of Data Integration* (Morgan Kaufmann, 2012).
4. Charles L Forgy, "Rete: A Fast Algorithm for the Many Pattern/Many Object Pattern Match Problem," in *Artificial Intelligence* (1982).

- **Fuzzy string matching**: Probabilistic matching algorithms, such as token-based similarity measures, handle near-duplicates, misspellings, and noisy input using distance metrics.[5] These are lightweight yet effective for aligning ERP field labels and categorical values.

- **Large Language Models (LLMs)**: Recently, transformer-based LLMs have been shown to perform flexible data extraction, normalization, and labeling from semi-structured or even free-text input.[6] Within ERP data pipelines, LLMs offer redundancy for failed deterministic or statistical mappings by leveraging contextual understanding.

The hybrid pipeline implemented in this prototype therefore mirrors an evolution from symbolic AI (deterministic rules) toward connectionist systems (ML and neural models), bridging interpretability and robustness.

4.3 Rule-Based Systems vs. Machine Learning

Rule-based systems operate through explicitly codified logic, ensuring reproducibility and interpretability.[7] However, as the diversity of data sources grows, purely rule-based architectures exhibit exponential growth in maintenance overhead. They fail to generalize to unseen schemas or contextual variations. Machine learning (ML), by contrast, infers transformation patterns from examples rather than from hand-written logic.[8] In ERP data integration, ML enables schema alignment and entity-linking even when semantic or lexical mismatches occur. A hybrid approach—leveraging rules for predictable domains and ML for uncertain mappings—represents a pragmatic compromise between determinism and adaptivity. This mixed architecture is consistent with current AI trends in enterprise systems that emphasize interpretable automation, model trans-

5. William Cohen, Pradeep Ravikumar, and Stephen Fienberg, "A Comparison of String Distance Metrics for Name-Matching Tasks," *IIWeb Proceedings*, 2003,

6. Tom B. Brown, Benjamin Mann, and Nick et al. Ryder, "Language Models are Few-Shot Learners," *Advances in Neural Information Processing Systems* 33 (2020): 1877–1901; Wayne Xin Zhao, Kun Zhou, and Junyi et al. Li, "A Survey of Large Language Models," *arXiv preprint arXiv:2303.18223*, 2023,

7. Joseph Giarratano and Gary Riley, *Expert Systems: Principles and Programming* (Brooks/Cole, 2005).

8. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, *Deep Learning* (MIT Press, 2016).

parency, and feedback-driven learning loops.[9]

4.4 Types of Demand Forecasting

Demand forecasting is a critical element in ERP-driven decision optimization. It aims to predict future product demand based on historical and contextual factors.

4.4.1 Time Series Analysis

Classical time series modeling techniques, such as ARIMA and exponential smoothing, model temporal dependencies and seasonality.[10] These methods are statistically efficient and well-suited for stationary patterns in inventory and sales data.

4.4.2 Causal Models

Causal or econometric models incorporate exogenous predictors (e.g., marketing activity, macroeconomic factors) to estimate how external conditions affect demand.[11] Such models suit enterprise planning tasks requiring interpretability and policy sensitivity.

4.4.3 Machine Learning Approaches

Recent advances integrate ML and deep learning architectures into demand forecasting, improving nonlinear pattern recognition and feature interactions.[12] Recurrent (LSTM/GRU) or transformer-based models capture long-term dependencies, outperforming traditional methods on complex datasets. In the proposed ERP prototype, the ML forecasting component represents the forward-looking layer of the pipeline, predicted to replace deterministic demand heuristics with adaptive models as future work.

9. David Gunning, "Explainable Artificial Intelligence (XAI)," *Defense Advanced Research Projects Agency (DARPA)*, 2017,

10. Rob J. Hyndman and George Athanasopoulos, *Forecasting: Principles and Practice* (OTexts, 2018).

11. James H. Stock and Mark W. Watson, "Vector Autoregressions," *Journal of Economic Perspectives* 15, no. 4 (2001): 101–115.

12. Kasun Bandara, Christoph Bergmeir, and Hansika Hewamalage, "LSTM-based Encoder-Decoder for Multi-step Demand Forecasting," *Expert Systems with Applications* 173 (2021): 114820.

5 System AI Connector Architecture

5.1 Functionality Overview

5.2 Pipeline Architecture and Data Flow

The architecture is structured as a multi-layered hybrid pipeline.
The execution flow is illustrated below:

```
HTTP Request (JSON/JSONL/CSV)
        |
        v
   AdapterFactory          <- content-type detection, normalization
        |
        v
   RuleEngine              <- deterministic ERP-specific field mapping
                              (6 ERP profiles + generic fallback)
        |
        v
   MLEngine                <- fuzzy value matching via RapidFuzz
        |
        v
   Dependency Synthesis    <- auto-generates Product -> Location ->
                              StorageLocation -> Storage -> Employee -> Request
        |
        v
   Validator (Pydantic)    <- schema validation against 8 DTO models
        |
        v
   LLMEngine (fallback)    <- Ollama/Qwen2.5-Coder for failed deterministic pa
        |
        v
   BackendClient           <- REST POST to Spring ERP, UUID translation,
                              idempotency handling
```

5.3 Layer 1: Data Normalization Layer (Adapter)

- Supported adapters: `JsonAdapter`, `JsonLinesAdapter`, `CsvAdapter`, `FreeTextAdapter`.
- `AdapterFactory` selects the appropriate adapter based on the Content-Type header.

5.4 Layer 2: Rule Engine

- 6 ERP-specific mapping dictionaries (Odoo, SAP S/4HANA, Epicor Kinetic, Oracle NetSuite, Microsoft Dynamics 365, Infor CloudSuite) plus a generic fallback.
- 10-step processing: envelope extraction → flattening → key mapping → date normalization → compound field splitting → storage name synthesis → type coercion → context-aware remapping → address flattening → volume derivation.
- Characteristics: Deterministic, zero-latency, fully testable.

5.5 Layer 3: ML Engine

- `RapidFuzz` fuzzy matching (`token_sort_ratio`, threshold 80) for value normalization (e.g., usage types, UoM, country codes).
- `scikit-learn` integration available for future model extensions (demand forecasting).
- Lightweight execution requiring no GPU.

5.6 Layer 4: Dependency Synthesis

- For `inventory` and `transfer` record types, the pipeline auto-generates the full entity dependency chain.
- Deterministic UUID generation (`string_to_uuid`) ensures idempotency across repeated imports.
- Transfer records additionally synthesize a system Employee and Request entity.

5.7 Layer 5: Validation

- Pydantic v2 models for 8 Data Transfer Objects (DTOs): `ProductCreateDto`, `LocationCreationDto`, `StorageLocationCreationDto`, `StorageCreationDto`, `InventoryCreationDto`, `RequestCreateDto`, `ShipmentCreateDto`, `EmployeeCreateDto`.
- `Validator.determine_model()` uses heuristic key-set matching combined with `record_type` hints.

5.8 Layer 6: LLM Fallback Engine

- Local Ollama instance running the `qwen2.5-coder:7b` model.
- Invoked exclusively when the rule engine and validation fail, thereby preserving overall system performance.
- Structured JSON output is enforced via Ollama's `format: json` parameter.
- Prompt engineering leverages schema descriptions alongside the raw record to ensure structured output.

5.9 Layer 7: Backend Integration

- `BackendClient` routes processed data to 8 distinct REST endpoints.
- A UUID translation map handles foreign-key references between sequentially created entities.
- Idempotency is maintained via duplicate detection through response body inspection, with a backend lookup fallback for ID recovery.

6 Design Justifications and Trade-offs

6.1 When Does Type-Agnostic ERP Parsing Work?

6.2 Impact of Fuzzy Matching on Accuracy

6.3 Local LLM Performance vs. Cloud API

- Data privacy considerations.
- Latency predictability.
- Cost and infrastructure constraints.

7 Evaluation

7.1 Test Datasets

- `erp_dataset.jsonl`, `erp_dataset_medium.jsonl`, `erp_dataset_small.jsonl`, `edge_cases.jsonl` — synthetic ERP records across all 6 supported systems.
- `test_report.csv` — pipeline run results.

7.2 Metrics

- Transformation success rate (rule/ML path vs. LLM fallback path).
- LLM invocation rate (where a lower rate indicates better rule coverage).
- Validation pass rate per DTO model.
- End-to-end latency (rule path vs. LLM path).
- Idempotency correctness (duplicate handling effectiveness).

7.3 Results

(To be filled from actual test_report.csv data)

- Rule engine handles X% of records without LLM.
- LLM fallback recovers Y% of remaining records.
- Overall pipeline success rate: Z%.

7.4 Evaluation of Decision Quality

8 Discussion

8.1 Hybrid vs. Pure Rule-Based Performance

The hybrid approach consistently outperforms pure rule-based systems, especially when encountering unknown or irregular schemas.

8.2 Latency and Adaptability Trade-offs

While the LLM fallback introduces notable latency (approximately 30-second timeouts), it significantly improves system adaptability to edge cases.

8.3 System Limitations

- The LLM requires a local Ollama instance, increasing local resource overhead.
- The ML engine is currently limited to fuzzy matching; the PyTorch demand forecasting model is not yet deployed in the live prototype.

8.4 Future Work

- Integrate a PyTorch LSTM/Transformer model specifically for demand forecasting on inventory records.
- Implement active learning loops to dynamically improve rule coverage over time based on successful LLM fallbacks.

9 Conclusion

- The prototype successfully demonstrates that a layered AI pipeline (rules $\rightarrow$ ML $\rightarrow$ LLM) can automate ERP data integration with high accuracy and robust adaptability.
- The architecture proves highly extensible, with the `MLEngine` stub actively prepared for future PyTorch model integration.
- This research provides a practical, reusable framework for AI-driven ERP data harmonization.

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